MIL-B-14036D
24 April 1985
SUPERSEDING
MIL-B-14036C
12 March 1969

MILITARY SPECIFICATION

BRACKET, OUTBOARD MOTOR: BOW ATTACHMENT; 12-TON, PNEUMATIC HALF FLOAT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a bracket for mounting outboard motors on the bow of a 12-ton pneumatic half float.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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SPECIFICATIONS

FEDERAL

PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein.

DRAWINGS

TA13214E9400	- Bracket, Outboard Motor: Bow Attachment, 12-Ton Pneumatic Half Float.

(Copies of specifications, standards, and drawings required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

ANSI/AWS D1.1, Structural Welding Code - Steel, Section 5, Qualification.

(Applications for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The outboard motor bracket (hereinafter referred to as "bracket") shall consist of a frame, transom, puller, and platform and shall be as shown on TA13214E9400, and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the bracket. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

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3.3 Material. Material shall be as specified herein and as shown on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The bracket shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar Metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming the MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the bracket may be newly fabricated from recovered materials to the maximum extent practicable, provided the bracket produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the bracket.

3.4 Puller assembly. The puller assembly shall include a brake, clutch, and a ratchet with pawl. The puller shall be capable of moving a cable load of 600 pounds without damage, permanent deformation, or failure of any part. The puller shall withstand a load of 2,000 pounds without permanent deformation, failure of any component, or release of the winch mechanism. Directions for lifting and lowering shall be permanently marked on the puller and shall be visible.

3.5 Identification marking. The bracket shall be identified in accordance with MIL-STD-130 and as shown on the drawings.

3.6 Treatment and painting. The bracket shall be cleaned, treated, and painted in accordance with MIL-T-704, type B.

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3.6.1 Cleaning. All ferrous metal surfaces requiring painting shall be cleaned to the bare metal by sandblasting to remove all oil, grease, mill scale, products of erosion, dirt, welding slag or spatter, and other foreign substances that may be detrimental to the coating system.

3.7 Workmanship. All parts, components, and assemblies of the bracket including welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful, extraneous material. External surfaces shall be free from burns, sharp edges, and corners except when sharp edges and corners are required.

3.7.1 Steel fabrication. Steel used in the fabrication of the equipment shall provide original quality surface finish and shall be free from kinks and sharp bends. Steel having an eroded surface is not acceptable. The forming of material shall be done by methods that will not cause damage to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Burned surfaces of flame-cut material shall be free of slag. Precautions shall be taken to avoid overheating, and heated metals should be allowed to cool slowly except where heat treatment is required. All bends of a major character shall be made with metal dies or jigs to insure uniformity of size and shape.

3.7.2 Plywood fabrication. Plywood shall be drilled and cut square and true using cutting blades, speeds, and feeds that will not split or splinter the corners and outside veneers. Sharp edges that are exposed in the assembled bracket shall be rounded and sanded smooth. Wood screws shall not be hammer-driven. Pilot holes for wood screws shall be centered and shall not exceed 90 percent of the length of the screw. Diameter of the lead holes shall not exceed 70 percent of the root diameter of the screw.

3.7.3 Welders and welding. Before assigning any welder to manual welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1, Structural Welding Code-Steel, Section 5, Qualification.

Certification shall be made available for review by the contracting officer or designated representitive. Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with AWS or ASME codes.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the first produced bracket shall be cause for disassembly, in the presence of a Government representative, of the first produced bracket to the extent necessary to determine the cause of failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first produced bracket. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.5).

4.2 Classification of inspections.

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article bracket shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.3.2 Tests. The first article bracket shall be tested as specified in 4.5.2. Failure of any test shall be cause for performing the inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Examination. Each bracket shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.2 Tests.

4.4.2.1 Individual. Each bracket shall be tested as specified in 4.5.2. Failure of this test shall be cause for rejection.

4.5 Inspection procedures.

4.5.1 Examination. The brackets shall be examined as specified herein for the following defects:

Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatment.
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the bracket.
106. Bracket not true to dimensions and lines, contains twists, bends, or other imperfections.
107. Assembly incorrect.
108. Components missing or not as specified.
109. Chains not as specified.
110. Any parts or components not in accordance with the QAP requirements as shown on the drawings.
111. Marking illegible, incorrect, incomplete, or missing.
112. Treatment and painting not as specified.
113. Improper finishing of hardware.
114. Workmanship not as specified.
115. Steel fabrication not as specified.
116. Plywood fabrication not as specified.
117. Welder certification not available for review.
118. Welding and weldments not in accordance with requirements (see D13214E9223).

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4.5.2 Tests.

4.5.2.1 Test procedure. The bracket shall be mounted on a test stand in a manner duplicating the mounting on Drawing D13214E1400-1. A load of 600 pounds shall be placed on the operator's platform. The transom board assembly shall be winched the full length of its travel in both directions not less than five times. The brake shall be engaged during the raising and lowering operation. The platform shall be centered in its travel, and the transom board loaded with an additional load of 1400 pounds. Failure to conform with 3.4 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective:

119.	Materials and containers not as specified for level A or B. Each incorrect material or container shall be considered one defect.
120.	Disassembly not as specified for level A.
121.	Unprotected metal surfaces not coated with preservative for level A.
122.	Blocking and bracing not in accordance with referenced document.
123.	Strapping not finish B for level A.
124.	Marking missing, illegible, incorrect, or incomplete for level A, level B, or Commercial.

5. PACKAGING

5.1. Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The bracket shall be disassembled to the maximum practical extent consistent with effecting a reduction in cube while providing ease of packing and preventing damage during shipment.

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5.1.1.2 Unprotected surfaces. Unpainted or otherwise unprotected metal surfaces requiring a contact preservative in accordance with the criteria of MIL-P-116 shall be coated with P-1 preservative as specified in MIL-P-116.

5.1.2 Commercial. The bracket shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. The components comprising a complete bracket shall be packed in a close fitting box conforming to PPP-B-621, class 2. Blocking and bracing necessary to prevent movement and damage shall be in accordance with MIL-STD-1186. Strapping shall be in accordance with the appendix to the box specification and unless otherwise specified (see 6.2), shall be finish B.

5.2.2 Level B. The components comprising a complete bracket shall be packed as specified in 5.2.1 except the box shall be class 1 and strapping is not required to be finish B.

5.2.3 Commercial. Each bracket shall be packed in accordance with ASTM D 3951.

5.3 Marking.

5.3.1 Military. Marking for level A and B preservation and packing shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial preservation and packing shall be in accordance with ASTM D 3951. Additionally, each unit shall be marked with the gross weight and cube.

6. NOTES

6.1 Intended use. The bracket is intended for use on the bow of 12 ton pneumatic half floats, for mounting one or two 25 horsepower outboard motors.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval, and the number of units required (see 3.2 and 6.3).

c. Level of preservation and level of packing required (see 5.1 and 5.2).

d. When other than finish B strapping is required (see 5.2.1).

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6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of 1 or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the documents first article.

6.4 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.5 Definitions.

6.5.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-14036D	Bracket, Outboard Motor: Bow Attachment: 12-Ton, Pneumatic Half Float

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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